

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5 HT	(a)			Qualitative involves identifying the presence of substances (1) Quantitative involves measuring /how much of a substance/masses/volumes/ amounts / concentrations of substances (1)	2			2		
	(b)	(i)		Calcium & magnesium (1) Accept $\text{Ca}^{2+}/\text{Ca}^{2+}/\text{Mg}^{2+}/\text{Mg}^{2+}$			1	1		1
		(ii)		Sulfate & carbonate & chloride (1) Accept $\text{SO}_4^{2-}/\text{SO}_4^{2-}/\text{SO}_4^{2-}/\text{CO}_3^{2-}/\text{CO}_3^{2-}/\text{CO}_3^{2-}/\text{Cl}^-/\text{Cl}^-$			1	1		1
	(c)			Flame test (1) Compare colour of flame (against a chart) (1)	2			2		2
	(d)			Calcium sulfate/carbonate/chloride (1) $\text{CaSO}_4/\text{CaCO}_3/\text{CaCl}_2$ (1) OR magnesium sulfate/carbonate/chloride (1) $\text{MgSO}_4/\text{MgCO}_3/\text{MgCl}_2$ (1) allow ecf from (b)			2	2		2
				Question 5 total 8	4	0	4	8	0	6